

PROJECT REPORT

Analysing Mental Health in Student Ecosystem

A Tableau-based Data Analysis and Visualization Project

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Course	B.Com (Hons.)
Project Area	Data Analysis & Visualization using Tableau

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Declaration

I, Saksham Yadav, student of B.Com (Hons.) at IITM Janakpuri, declare that this project report titled “Analysing Mental Health in Student Ecosystem” has been prepared as part of my academic project work. The dashboard and analysis presented here were developed using Tableau for the purpose of understanding patterns in the given student dataset. The observations are for academic and analytical use and are not intended to provide medical diagnosis.

Acknowledgement

I would like to thank my college and faculty for providing the opportunity to work on a practical data visualization project. This project helped me understand how raw data can be converted into visual information that is easier to interpret. I also gained practical experience in Tableau, including creating charts, applying measures and dimensions, and arranging individual worksheets into a dashboard.

1. Abstract

Mental health has become an important part of the student experience because academic pressure, sleep patterns, study routines and social factors can affect how students feel and perform. This project uses a student-level dataset to explore selected indicators related to mental health and academic life. The analysis was carried out in Tableau and presented through an interactive dashboard.

The dashboard contains 1,000 student records and brings together measures such as stress level, sleep, study hours, academic performance index, depression level, gender and heart rate variability observations. The purpose is not to prove medical relationships, but to demonstrate how data visualization can help identify patterns, compare groups and communicate findings clearly.

2. Introduction

Student life involves academic work, personal responsibilities, changing routines and social pressure. Because several factors can operate at the same time, a dataset containing student information can be useful for exploring broad patterns. In this project, Tableau was used to convert the available data into individual visualizations and then combine them into a single dashboard. The result gives a quick overview while also allowing each chart to be examined separately.

3. Objectives of the Project

- To study selected mental-health-related indicators in a student dataset.
- To compare stress and depression-related measures across gender groups.
- To examine the distribution of students by mental health risk-related records.
- To observe the pattern between study hours and the academic performance index.
- To visualize sleep, stress, depression and heart-rate-variability-related information.
- To build a clear Tableau dashboard that presents the major findings in one place.
- To develop practical skills in data visualization and dashboard design.

4. Scope of the Study

The scope is limited to the variables available in the supplied student dataset. The analysis is descriptive rather than clinical. The dashboard can show patterns in the sample, but it cannot establish that one factor directly causes another.

5. Tools and Technologies Used

Tableau was used for data visualization, worksheet creation and dashboard design. The supplied student dataset was used as the source. Microsoft Word was used to prepare this report.

6. Data and Methodology

The project followed a straightforward workflow: load and inspect the data in Tableau, identify relevant fields, create individual worksheets, apply suitable aggregations such as AVG, SUM and COUNT, compare groups, arrange the worksheets into a dashboard, and review the final visualizations for readability.

- Load and inspect the student data in Tableau.
- Identify variables useful for the project objectives.
- Create individual worksheets for the required views.
- Apply appropriate aggregation methods such as average, sum and count.
- Compare gender groups using charts and a treemap.
- Create the study-hours versus academic-performance visualization.
- Arrange the worksheets into a single dashboard.

7. Key Variables Used

Variable / Measure	Use in Dashboard	Visualization / Summary
Gender	Group comparison	Treemap, bar chart, pie chart
Stress Level	Overall and gender comparison	Average / treemap
Sleep	Overall sleep indicator	Dashboard KPI
Study Hours Per Day	Academic routine	X-axis of performance chart
Academic Performance Index	Academic outcome indicator	Y-axis of performance chart
Depression Level	Depression-related measure	Gender comparison / total
Mental Health Risk	Risk distribution	Gender-wise count
Heart Rate Variability	Physiological indicator	Gender-wise observation count

8. Tableau Dashboard

The final dashboard was designed to provide an overall snapshot of the student ecosystem. The top section highlights key figures, while the remaining sections compare stress, mental health risk, heart rate variability, study hours, academic performance and depression.

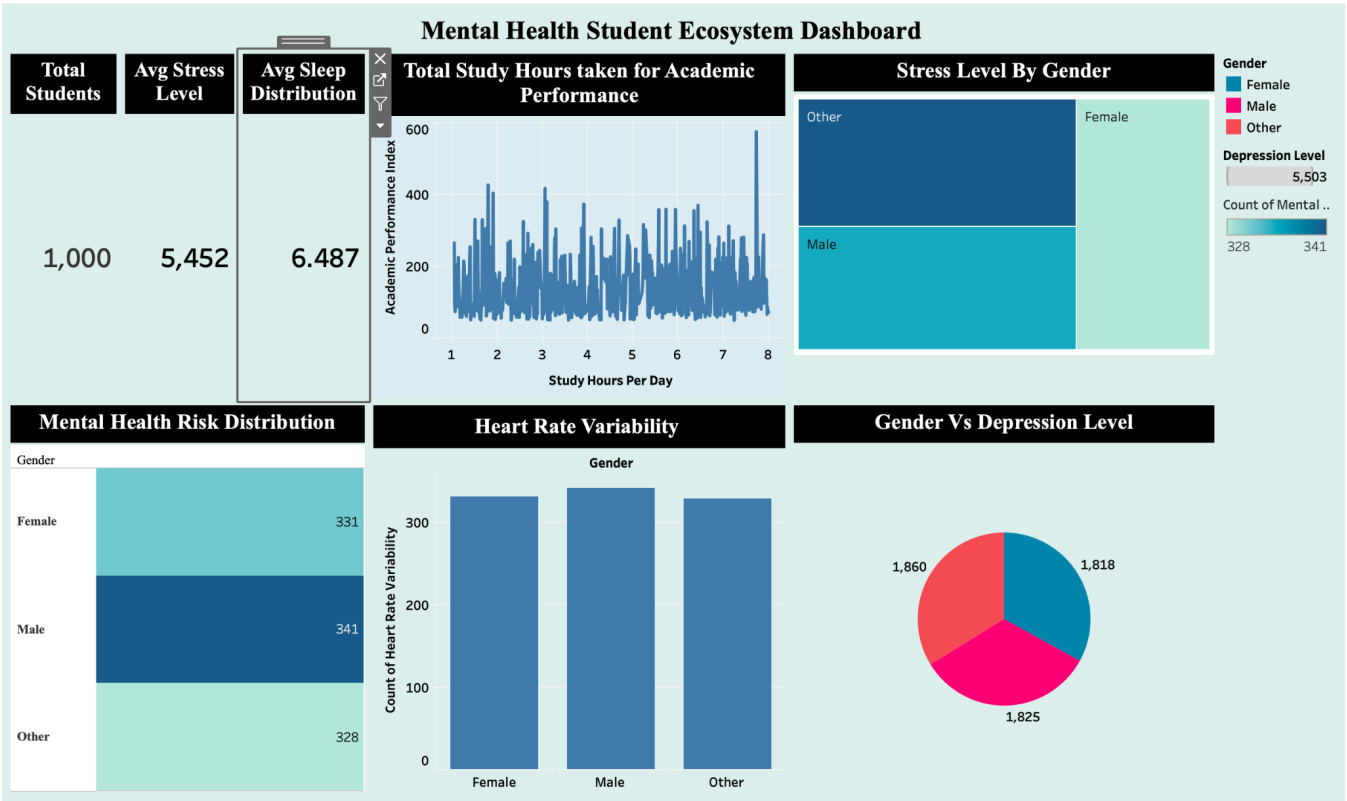


Figure 1: Final Tableau dashboard – Analysing Mental Health in Student Ecosystem

9.1 Mental Health Risk Distribution

The displayed counts are 331 for Female, 341 for Male and 328 for Other. The values are close to one another, indicating that the dataset is fairly balanced across the three gender categories.

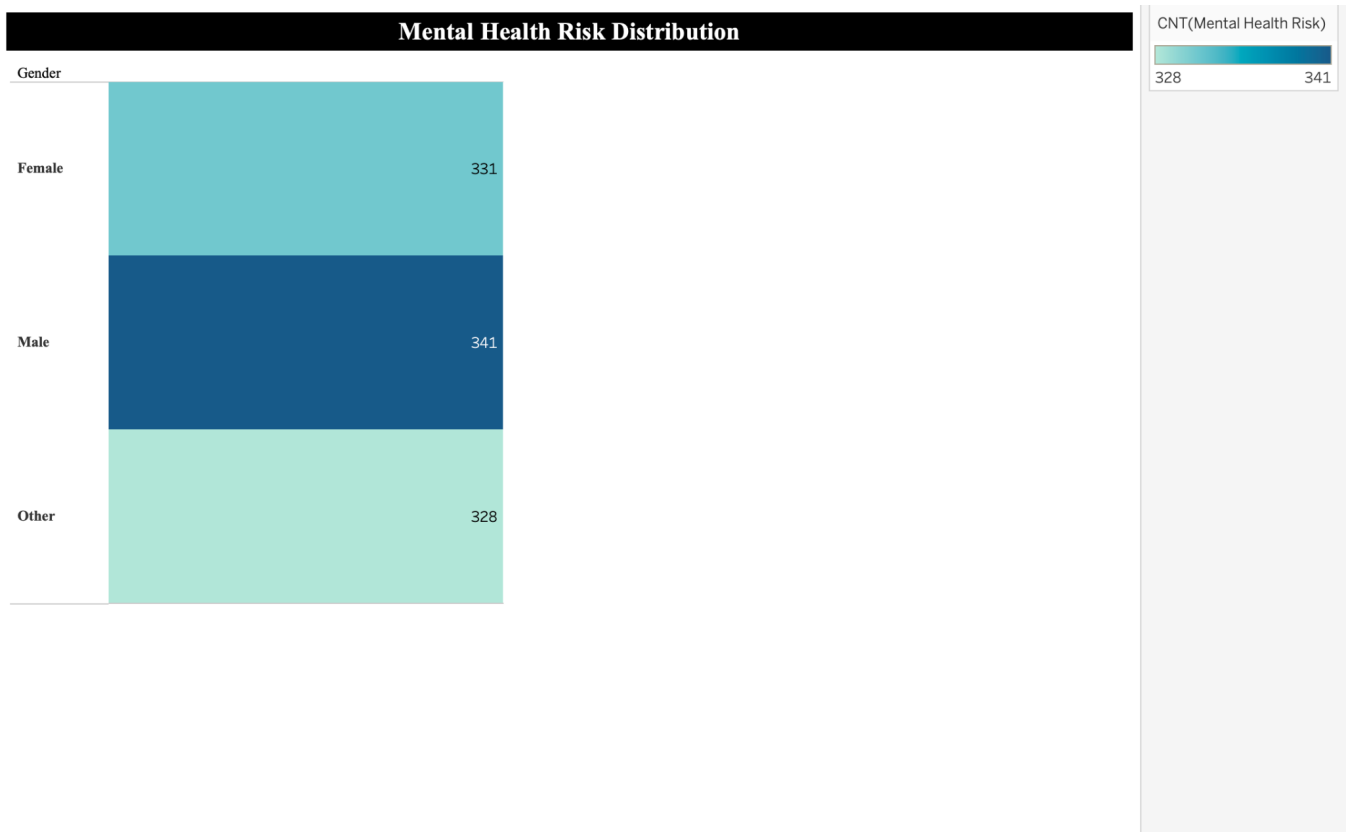


Figure 2: Mental Health Risk Distribution

9.2 Heart Rate Variability

The bars are close in height, with the Male category slightly higher than Female and Other. The worksheet uses COUNT on the vertical axis, so it should be read as a count of recorded observations rather than an average HRV value.

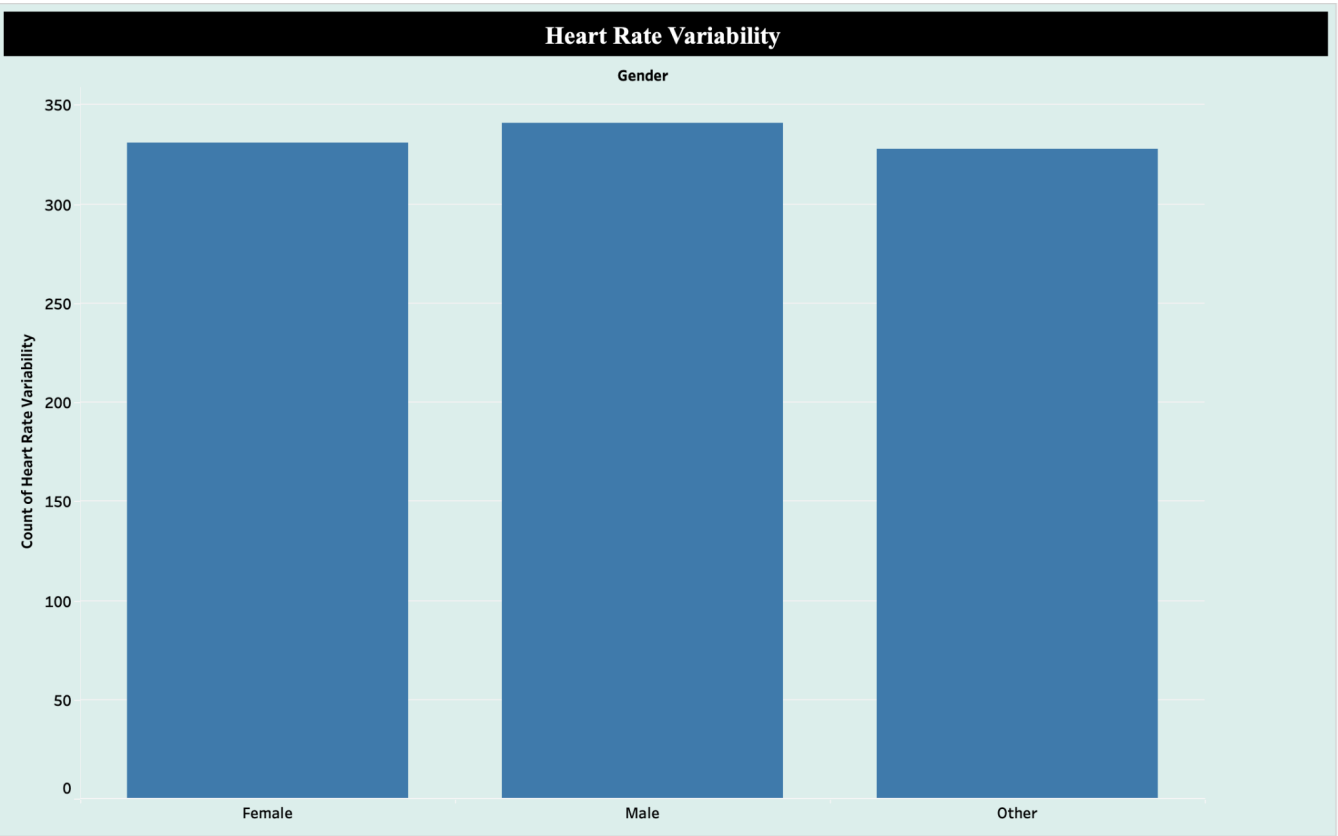


Figure 3: Heart Rate Variability by Gender

9.3 Stress Level by Gender

The displayed average stress values range from approximately 5.30 to 5.61. The differences are relatively small, so the visual should be interpreted as a close comparison rather than a large gender gap.

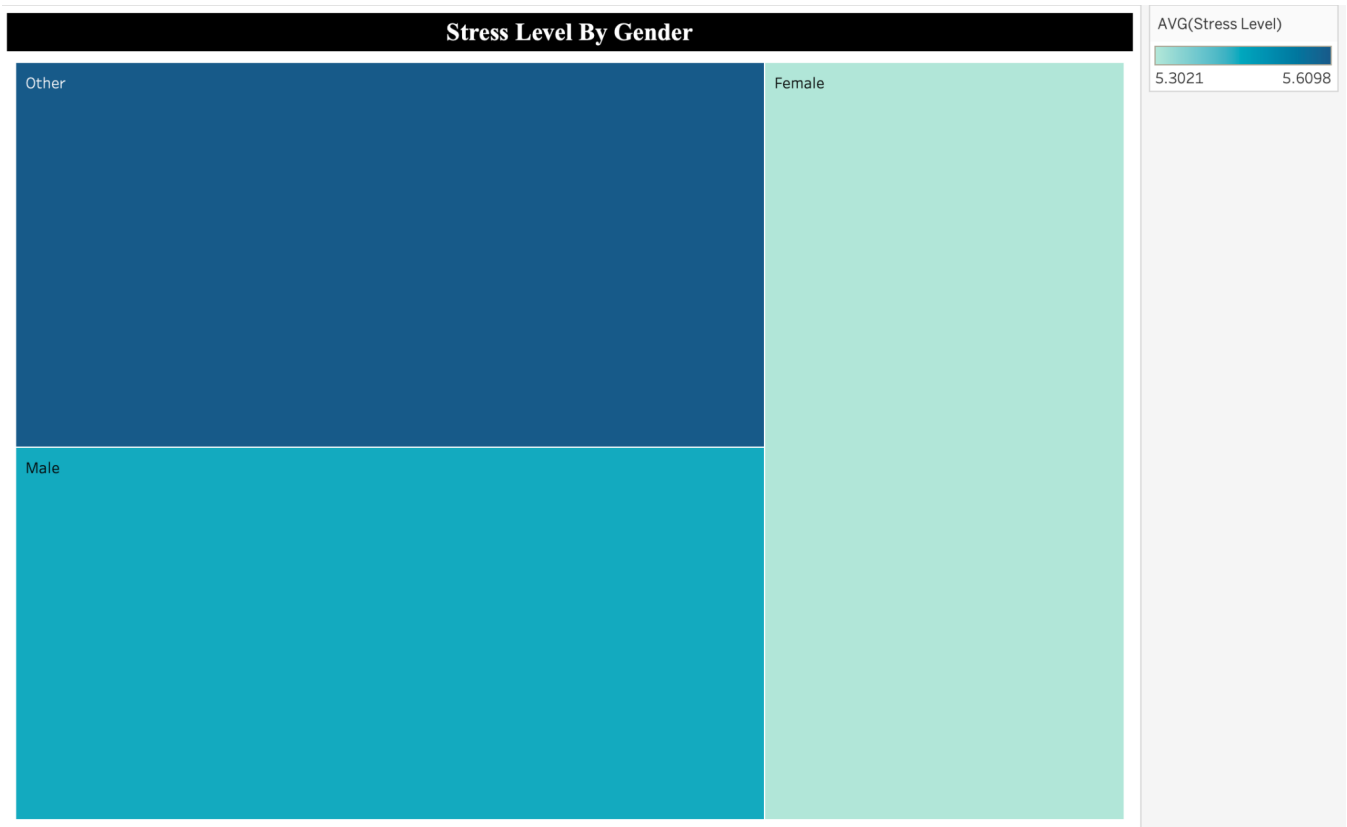


Figure 4: Stress Level by Gender

9.4 Study Hours and Academic Performance

The chart shows considerable variation in the Academic Performance Index at different study-hour values. Several high and low observations appear across the range, suggesting that study time alone does not explain all of the variation in academic performance.

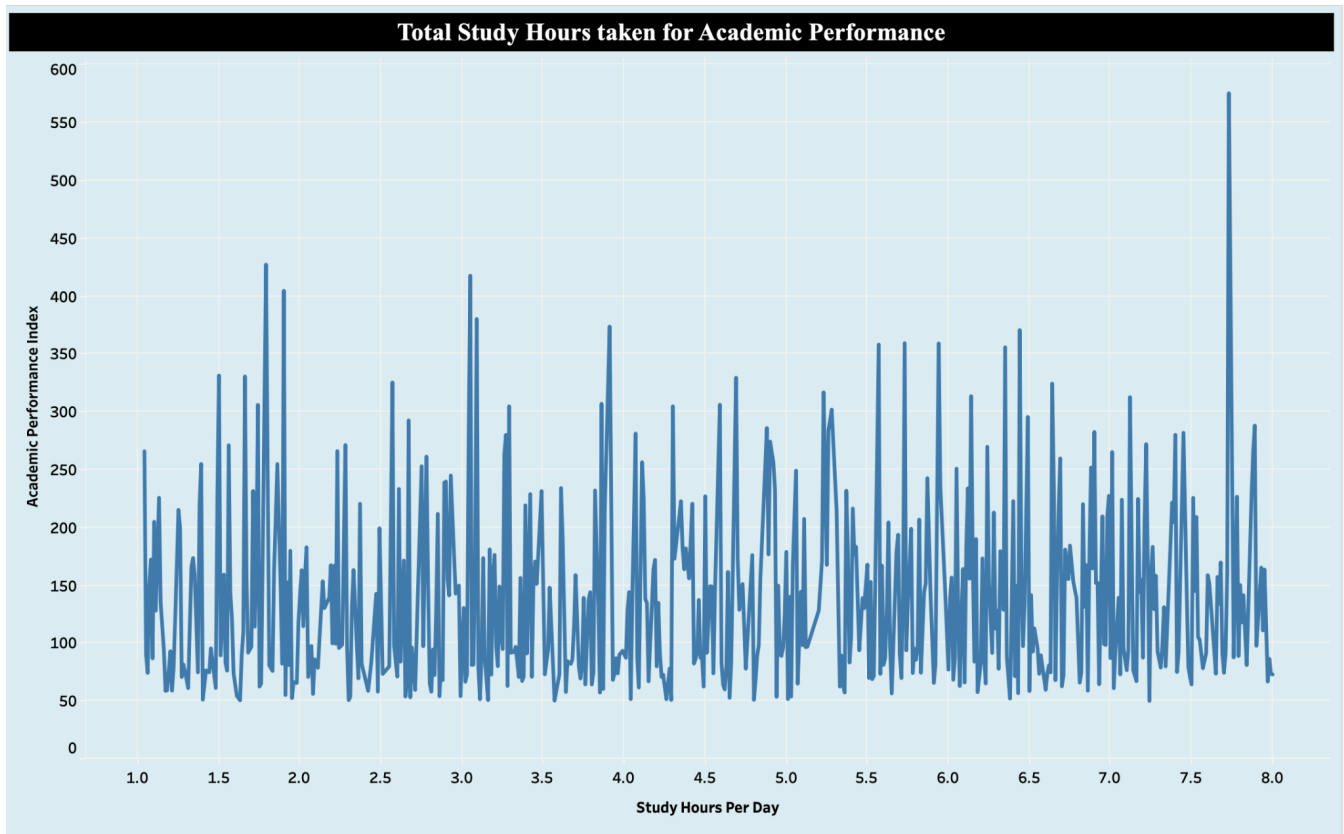


Figure 5: Study Hours Per Day vs Academic Performance Index

9.5 Gender vs Depression Level

The pie chart displays the SUM of Depression Level by gender: 1,818 for Female, 1,825 for Male and 1,860 for Other, giving a displayed total of 5,503. The three totals are relatively close.

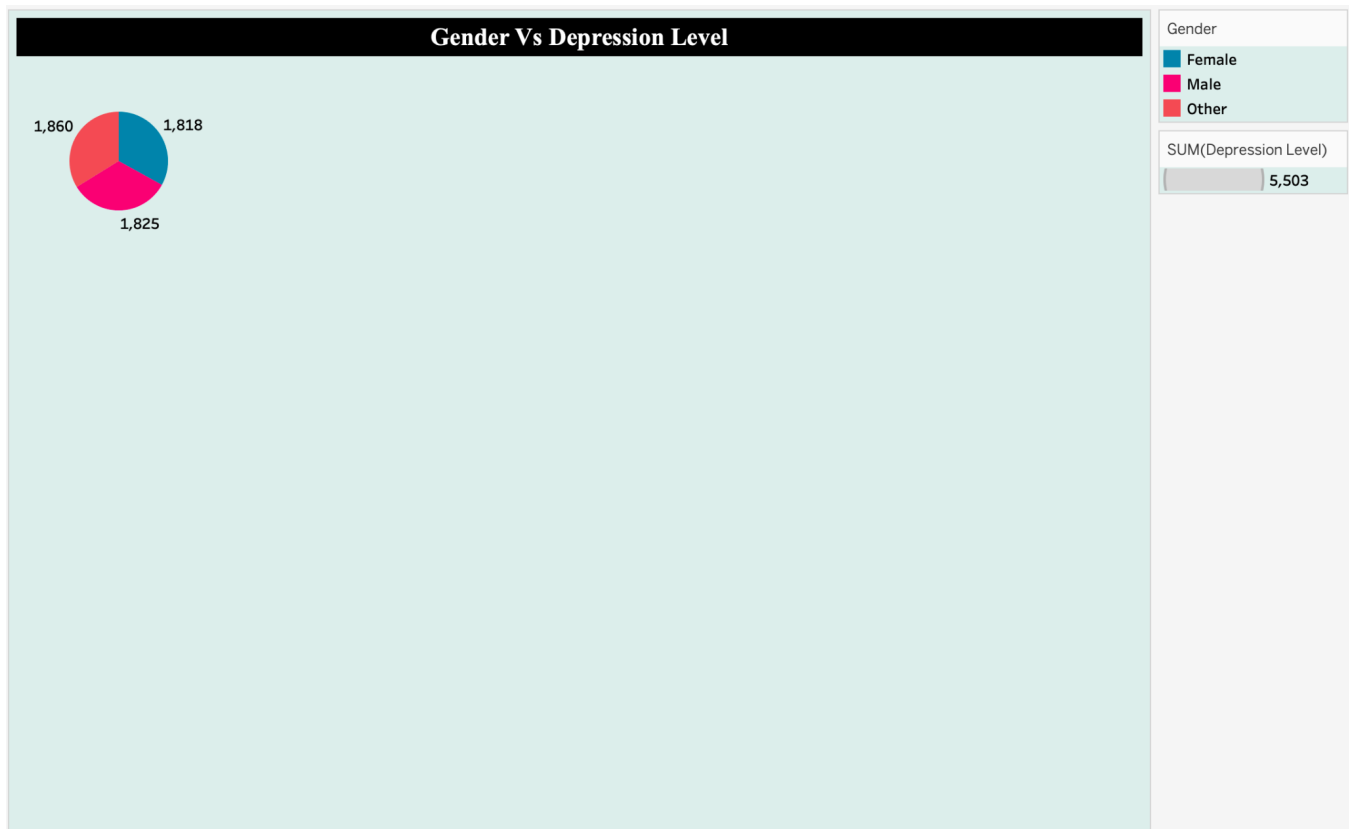


Figure 6: Gender vs Depression Level

10. Key Dashboard Findings

- The dataset contains 1,000 student records.
- The dashboard shows an average stress level of 5.452.
- The displayed average sleep value is 6.487.
- Mental health risk-related record counts are 331 Female, 341 Male and 328 Other.
- Gender-wise mental health risk counts are very close, suggesting a fairly even sample distribution.
- The gender-wise heart-rate-variability observation counts are also close.
- The displayed depression-level total is 5,503, split across Female (1,818), Male (1,825) and Other (1,860).
- The study-hours/performance chart shows substantial variation, so study hours alone should not be treated as a complete explanation for academic performance.

11. Interpretation of the Results

The dashboard is useful because it brings academic and well-being indicators together. The number of records across gender categories is fairly even, making group comparisons easier. The stress-related view shows averages within a relatively narrow range, while the depression totals are also close when viewed using the selected SUM measure. These differences should therefore be discussed carefully rather than presented as major gender-based differences.

The study-hours chart is another useful part of the dashboard because it shows how varied student outcomes

can be. Students with similar study hours do not necessarily have the same academic performance index. This suggests that academic performance is influenced by more than study time alone.

12. Limitations

- The analysis is based only on the variables available in the provided dataset.
- The dashboard is descriptive and does not establish cause-and-effect relationships.
- Aggregated values can hide individual student-level differences.
- The HRV chart uses a count of observations and should not be interpreted as an average physiological value.
- Mental health indicators in a dataset cannot replace professional psychological or medical assessment.
- Qualitative factors such as interviews, family circumstances and student feedback are not included.

13. Future Scope

- Add filters for age, academic year, course or other available characteristics.
- Use scatter plots and trend lines to examine relationships more directly.
- Include correlation analysis between stress, sleep, study hours, depression and academic performance.
- Add time-based data if repeated observations are available.
- Develop a privacy-conscious risk-monitoring view for institutions.
- Improve the dashboard with interactive tooltips and drill-down views.

14. Conclusion

This project demonstrates how Tableau can be used to turn a student dataset into a practical visual report. The final dashboard gives a compact view of stress, sleep, mental health risk, depression, study hours, academic performance and heart-rate-variability-related information.

The analysis also shows why student well-being should not be represented by a single variable. Different indicators need to be considered together, and visualization makes it easier to compare them and notice patterns. Overall, the project provided useful hands-on experience in data analysis, visualization, dashboard creation and communicating findings.

15. References / Sources

- Student dataset provided for the academic project.
- Tableau – data visualization and dashboard development software.
- Tableau official documentation and learning resources for charts, measures and dashboard features.

— End of Project Report —